Roth's Theorem — The First Density Theorem

If van der Waerden says any coloring of $\mathbb{N}$ forces a 3-term AP, Roth says you don't even need to color everything. A single set occupying positive fraction of integers already forces a 3-term AP.

First time density, not partition, forces arithmetic order — $k = 3$ case of Szemerédi, analytic heart that later powers Green-Tao.

> **Roth's Theorem 1953:** Any subset of $\mathbb{N}$ with positive upper density contains a non-trivial 3-term AP.

Proved by Klaus Roth — Fields Medal 1958 — launched modern additive combinatorics.

Formal Statement

Upper density:

$$\bar{d}(A) = \limsup_{N\to\infty} \frac{|A \cap [1, N]|}{N}$$

Fraction of $\{1, \ldots, N\}$ belonging to A, taken $\limsup$ — best infinitely often.

If $\bar{d}(A) = \delta > 0$ then exists a, d with $d \neq 0$ and $a,\ a+d,\ a+2d \in A$ three equally spaced numbers inside A, non-trivial $d \neq 0$.

Note $\bar{d}$ uses $\limsup$, not limit — allows sets like numbers starting with 1, density oscillates 10% to 55%, still $\bar{d} > 0$ so Roth applies.

Quantitative finite form:

Let

$$r_3(N) = \max\{|A| : A \subset [1, N],\ A \text{ contains no non-trivial 3-AP}\}$$

Largest AP-free size.

Roth proved:

$$r_3(N) = O\left(\frac{N}{\log\log N}\right)$$

Because $\dfrac{1}{\log\log N} \to 0$, we have $\dfrac{r_3(N)}{N} \to 0$.

Consequence: Fix $\delta > 0$. For large N,

$$\delta N > C \cdot \frac{N}{\log\log N} \geq r_3(N)$$

since $\dfrac{1}{\log\log N} \to 0$. So any $A \subset [1, N]$ with $|A| \geq \delta N$ must contain 3-AP.

Take $A \subset \mathbb{N}$ with $\bar{d}(A) = \delta > 0$ — then infinitely many N with $|A \cap [1, N]| \geq \frac{\delta}{2}N$. For large such N, $|A \cap [1, N]| > r_3(N)$ $\to$ contains 3-AP. So infinite set with positive upper density contains 3-AP — actually infinitely many.

Question since: how fast does $\dfrac{r_3(N)}{N} \to 0$?

Roth gave $O\left(\frac{1}{\log\log N}\right)$. Behrend gives lower bound $\Omega\left(\exp\left(-C\sqrt{\log N}\right)\right) = \dfrac{1}{\exp\left(C\sqrt{\log N}\right)}$. Gap huge: $\exp(-C\sqrt{\log N})$ decays slower than any $(\log N)^{-c}$? Actually faster than $(\log\log N)^{-1}$? Let's compare: $\exp(C\sqrt{\log N}) \gg \log\log N$ for large N, so Behrend upper on density $\dfrac{r_3(N)}{N} \geq \dfrac{1}{\exp(C\sqrt{\log N})}$ much larger than Roth's $\dfrac{1}{\log\log N}$? No, $\dfrac{1}{\exp(\sqrt{\log N})} \ll \dfrac{1}{\log\log N}$ eventually — so Roth upper bound far from Behrend lower bound. Closing gap is 70-year program culminating in Bloom-Sisask $\dfrac{1}{(\log N)^{1+c}}$.

In van der Waerden language: $W(2,3) = 9$ says 2-coloring of $[1,9]$ forces monochrome 3-AP. Roth says you don't need to color all numbers — single color class of density $\delta > 0$ already forces 3-AP inside itself. Density version strictly stronger than coloring version for $k = 3$.

Why It's Hard: Naive Count Fails

Let $A \subset [1,N]$, $|A| = \delta N$. Count 3-APs including trivial $d = 0$:

$$T_3(A) = \sum_{x,d} 1_A(x)1_A(x+d)1_A(x+2d) = N^2 \sum_r \left|\widehat{1_A}(r)\right|^2 \widehat{1_A}(-2r)$$

Fourier identity.

What is T_3? T_3 counts pairs (x,d) with $x, x+d, x+2d \in A$. For each $a = x$, d arbitrary $(-N \leq d \leq N)$, includes $d = 0$ gives N trivial APs a, a, a. Nontrivial when $d \neq 0$. N^2 choices (x,d) total.

If A random density δ: Choose each $n \in [1,N]$ independently with probability δ. Then $\mathbb{E}\left[1_A(x)1_A(x+d)1_A(x+2d)\right] = \delta^3$ for $d \neq 0$ distinct points. So

$$\mathbb{E}\left[T_3(A)\right] \approx \delta^3 N^2$$

Many — N^2 vs N — so random set expects $\approx \delta^3 N^2$ progressions, e.g., $\delta = \frac{1}{10}$, $N = 10^6$ expects $10^{12} \cdot 10^{-3} = 10^9$ 3-APs. So random dense set has many APs.

Why naive expectation not proof? Structured set could conspire to cancel.

Example: evens only — still many APs, not counterexample. Need set where Fourier phases cancel to make T_3 small despite large δ.

Write balanced function:

$$f = 1_A - \delta 1_{[1,N]}$$

mean zero, $\hat{f}(0) = 0$.

Expand:

$$T_3(A) = \sum_{x,d} (\delta + f(x))(\delta + f(x+d))(\delta + f(x+2d))$$

Main term $\delta^3 \sum_{x,d} 1 = \delta^3 N^2$.

Cross terms involve $\sum_{r \neq 0} \left|\hat{f}(r)\right|^2 \hat{f}(-2r)$ etc. So

$$T_3(A) = \delta^3 N^2 + N^2 \sum_{r \neq 0} \left|\widehat{1_A}(r)\right|^2 \widehat{1_A}(-2r) + \text{lower}$$

If A has no nontrivial 3-AP, then $T_3(A) = N$ trivial only — essentially 0 compared to $\delta^3 N^2$. So sum over $r \neq 0$ must be $\approx -\delta^3 N^2$ to cancel main term.

Hence some $\left|\hat{f}(r)\right|$ must be large — cannot have all $\left|\hat{f}(r)\right|$ small.

Formally:

$$\exists r \neq 0 : \ \left|\hat{f}(r)\right| \gg \delta^2$$

with $\hat{f}(r) = \dfrac{1}{N} \displaystyle\sum_{x=1}^{N} f(x) e^{-2\pi i r x/N}$.

Plain meaning: Large Fourier coefficient at frequency r means A correlates with linear phase $e^{2\pi i r x/N}$ — wave with period N/r.

$$\left|\sum_{x \in A} e^{2\pi i r x/N}\right| \gg \delta^2 N$$

Random set has $\left|\displaystyle\sum_{x \in A} e^{2\pi i r x/N}\right| \approx \sqrt{\delta N} \ll \delta^2 N$ — flat spectrum, all frequencies small $O(\sqrt{N})$.

No 3-APs forces spike: some $r \neq 0$ where sum large — bias. A prefers regions where phase ≈ 1, avoids where ≈ -1.

Think: sprinkle dots randomly on circle — no direction stands out. If dots cluster in arc, Fourier at corresponding frequency spikes.

So strategy: no APs $\implies$ bias $\implies$ denser on subprogression $\implies$ iterate — Roth density increment.

This is why Fourier needed — counting alone $\delta^3 N^2$ heuristic says many APs, but need to rule out conspiratorial cancellation. Fourier converts cancellation into structural bias that can be exploited.

The Proof: Density Increment via Fourier — Plain Terms

Three ideas now standard:

Idea 1: Fourier bias.

Let $f = 1_A - \delta 1_{[1,N]}$ balanced function — mean zero, positive where A denser than average, negative where sparser.

If A has no nontrivial 3-AP, then $T_3(A) \approx 0$ ignoring trivial N APs with $d = 0$, but main term from average δ contributes $\delta^3 N^2$. To cancel $\delta^3 N^2$ to near 0, some Fourier coefficient must be large:

$$\exists r \neq 0 : \ |\hat{f}(r)| \gg \delta^2.$$

Large Fourier coefficient means A correlates with linear phase $e^{2\pi i r x/N}$ — wave. A prefers places where wave phase near 1, avoids where phase -1. Not uniform — clusters in arithmetic sense, e.g., prefers certain residues mod q.

Analogy: If you sprinkle dots randomly on circle, no direction stands out. If Fourier spike, dots cluster in some arc — bias.

Idea 2: From bias to density increment.

Large Fourier coefficient implies A significantly denser on some long arithmetic progression $P \subset [1, N]$ length $N' \geq N^c$.

Precisely: exists P, $|P| \geq N^c$ with

$$\frac{|A \cap P|}{|P|} \geq \delta + c'\delta^2$$

Gain $\Omega(\delta^2)$ density by restricting to P.

Why? $e^{\frac{2\pi i r x}{N}}$ roughly constant on progression with step q where $\frac{rq}{N}$ small mod 1. Pigeonhole over q — Dirichlet. Densest subprogression where phase near 1 must have higher density.

Intuition: If A clusters mod q, look at densest residue class mod q — it has higher density than average. If A clusters on intervals, look at densest interval.

Idea 3: Iterate to contradiction.

Now repeat. $A_1 = A \cap P$, rescaled to $[1, N']$. Has density $\delta_1 \geq \delta + c'\delta^2$ and still no 3-AP — subset of AP-free set still AP-free.

Apply again: get P_2, density $\delta_2 \geq \delta_1 + c'\delta_1^2$, etc.

Density increases by at least $\Omega(\delta^2)$ each step, so after $O\left(\frac{1}{\delta}\right)$ steps density would exceed 1 — impossible, since density ≤ 1.

Therefore process must stop because we found 3-AP before that.

$O\left(\frac{1}{\delta}\right)$ steps, each shrinking N to about $\sqrt{N}$ (or N^c), yields N must be at least $\exp\left(\exp\left(O\left(\frac{1}{\delta}\right)\right)\right)$ to run that many steps, giving $r_3(N) \ll \frac{N}{\log \log N}$.

This density increment strategy — if no pattern, then denser on substructure, iterate — is template for Szemerédi, Gowers, Green-Tao.

Same meta-theorem: disorder budget finite. Budget = $\frac{1}{\delta}$. Each step without pattern spends budget by increasing density. Cannot increase forever.

How Large Can AP-Free Set Be? Bounds War

Let $r_3(N)$ be max size of 3-AP-free subset of $$ — no $a, a+d, a+2d$ with $d \neq 0$ inside.[1][N]

Roth proved $\frac{r_3(N)}{N} \to 0$. Question: how fast?

Lower bound — Behrend 1946: how to avoid

Construction:

1. Cube $\{0, \ldots, m-1\}^d$.
2. Sphere layer $S_R = \left\{x : \sum_{i=1}^{d} x_i^2 = R\right\}$.
3. No 3-AP on sphere: if $x + z = 2y$ and $\|x\|_2^2 = \|z\|_2^2 = \|y\|_2^2 = R$, then $\left\|\frac{x+z}{2}\right\|_2 < R$ unless $x = z$ by strict convexity. So distinct x, y, z in AP cannot all lie on same sphere.
4. Choose densest R — $|S_R| \geq \frac{m^d}{dm^2}$.

5. Map to integers $n = \sum_{i=1}^{d} x_i (2m)^{i-1}$ base $2m$ — no carries, so $x + z = 2y \iff n_x + n_z = 2n_y$. AP-free in cube $\implies$ AP-free in $\mathbb{Z}$.

Optimizing $d \approx \sqrt{\log N}$ gives:

$$r_3(N) \geq N \cdot \exp\left(-C\sqrt{\log N}\right) = \frac{N}{\exp\left(C\sqrt{\log N}\right)}$$

So $r_3(N) = N^{1-o(1)}$ — almost linear. For $N = 10^6$, $\frac{N}{\exp\left(C\sqrt{\log N}\right)} \approx \frac{N}{50}$ — still 2%, no 3-AP.

Elkin 2008:

$$r_3(N) \geq N \cdot \frac{\log^{1/4} N}{\exp\left(C\sqrt{\log N}\right)}$$

extra $\log^{1/4} N$ from better sphere packing.

Upper bounds — 70 years pushing $\log\log$ down to $\log$:

- **Roth 1953:**

$$r_3(N) \ll \frac{N}{\log\log N}$$

One large Fourier coefficient $\left|\hat{f}(r)\right| \gg \delta^2$ gives increment $\delta \to \delta + c'\delta^2$ on P, $|P| \geq N^c$. After $O\left(\frac{1}{\delta}\right)$ steps density > 1.

- **Szemerédi & Heath-Brown:**

$$r_3(N) \ll \frac{N}{(\log N)^c}$$

for some small $c > 0$ — Bohr sets $B(S, \rho) = \left\{x : \left\|\frac{rx}{N}\right\| < \rho \ \forall r \in S\right\}$ instead of progressions.

- **Bourgain 1999-2008:**

$$r_3(N) \ll N \cdot \frac{\sqrt{\log\log N}}{\sqrt{\log N}} \quad \text{and} \quad N \cdot \frac{(\log\log N)^2}{(\log N)^{2/3}}$$

refined restriction, almost-periodicity.

- **Sanders 2011:**

$$r_3(N) \ll N \cdot \frac{(\log\log N)^5}{\log N}$$

almost $\frac{N}{\log N}$ — Croot-Sisask lemma.

- **Bloom & Sisask 2020 breakthrough:**

$$r_3(N) \ll \frac{N}{(\log N)^{1+c}} \qquad c > 0$$

Broke $\frac{N}{\log N}$ barrier.

Why new? Old: one large coefficient $\implies$ increment $\delta \to \delta + \Omega(\delta^2)$. Need $O\left(\frac{1}{\delta}\right)$ steps.

New: many large coefficients, large spectrum $\Delta_\eta(f) = \left\{r : \left|\hat{f}(r)\right| \geq \eta\right\}$ has additive structure — $\Delta + \Delta$ small. Spectral boosting uses $\langle 1_A * 1_A, f\rangle$ to boost η. Almost-periodicity: exists measure μ on Bohr set with $\|1_A * \mu - 1_A\|_2$ small, so density increment $\delta \to \delta + \Omega(\delta)$ on much larger Bohr set — loses only $\exp\left(-O\left(\frac{1}{\delta}\right)\right)$ not N^c.

So $O\left(\log \frac{1}{\delta}\right)$ steps $\implies$ $(\log N)^{1+c}$.

Why $\frac{N}{\log N}$ threshold matters for Erdős:

Erdős conjecture: $\sum_{a \in A} \frac{1}{a} = \infty \implies A$ contains 3-AP.

If $|A \cap [1, N]| \approx \frac{N}{\log N}$ then $\sum_{a \leq N} \frac{1}{a} \sim \int^N \frac{d\,|A \cap [1,t]|}{t} \sim \frac{\log N}{\log N} \to \infty$ diverges — harmonic $\sum \frac{1}{n \log n}$ diverges.

If $|A \cap [1, N]| \ll \frac{N}{(\log N)^{1+c}}$ then $\sum \frac{1}{a} \ll \sum \frac{1}{n\,(\log n)^{1+c}} < \infty$ — integral $\int^\infty \frac{dx}{x\,(\log x)^{1+c}} = \frac{1}{c\,(\log x)^c} < \infty$.

Hence Bloom-Sisask: any 3-AP-free A has $\sum \frac{1}{a} < \infty$, so divergent reciprocal sum forces 3-AP — $k = 3$ case of Erdős.

For $k \geq 4$, best lower bound still $\frac{N}{\exp\left(C\sqrt{\log N}\right)}$, best upper $\frac{N}{(\log N)^c}$ — huge gap, requires U^k norms, nilsequences, far beyond.

Why Bloom-Sisask Matters: Erdős \$3,000 Conjecture

Erdős offered prizes:

> **Erdős Conjecture on APs:** If $A \subset \mathbb{N}$ satisfies $\sum_{a \in A} \frac{1}{a} = \infty$, then A contains k-term APs for every k.

Divergent reciprocal sum means not too sparse. Primes satisfy since $\sum \frac{1}{p} = \infty$ harmonic over primes diverges like $\log \log N$.

Connection to $r_3(N)$:

If $r_3(N) \approx \frac{N}{\log \log N}$, set can be 3-AP-free and still have divergent reciprocal sum, because $\sum \frac{1}{\left(\frac{N}{\log \log N}\right)^{-1}}$ diverges? Let's see: size of AP-free set at N about $\frac{N}{\log \log N}$ → density $\frac{N}{\log \log N}$ → sum of reciprocals $\sum \frac{1}{a}$ about $\int \frac{dN}{N \log \log N}$? Actually integral of density $\frac{dN}{N}$? For set with $|A \cap [1, N]| \sim \frac{N}{\log \log N}$, sum $\sum_{a \leq N} \frac{1}{a} \sim \int^N \left(\frac{1}{t}\right) d|A \cap [1, t]| \sim \frac{\log N}{\log \log N} \to \infty$ diverges. So Roth bound says nothing about Erdős.

If $r_3(N) \ll \frac{N}{(\log N)^{1+c}}$, then any 3-AP-free A has $\sum \frac{1}{a} < \infty$ because $\sum N^{-1}(\log N)^{-1-c}$ converges — integral test: $\int \frac{dx}{x(\log x)^{1+c}}$ converges for $c > 0$.

Bloom-Sisask therefore proved:

> $k = 3$ **case of Erdős conjecture true.** Any set with divergent reciprocal sum contains 3-term AP.

First unconditional progress on Erdős for any $k \geq 3$. $\frac{N}{\log N}$ was psychological barrier — exactly threshold where harmonic series switches from divergence to convergence.

For $k \geq 4$, Erdős conjecture remains wide open. Even $k = 4$ would require $r_4(N) \ll \frac{N}{(\log N)^{1+c}}$, far beyond current technology — best $r_4(N)$ bounds still $\frac{N}{(\log N)^c}$.

From Roth to Szemerédi to Green-Tao

- **Roth $k = 3$:** Fourier + density increment. Needs one large Fourier coefficient — linear phase $e^{2\pi i r x/N}$.
- **Szemerédi $k \geq 4$, 1975:** Roth Fourier fails — 4-APs need quadratic Fourier. Szemerédi used pure combinatorics, invented Regularity Lemma — partitions graph into random-like pieces. Vastly more complex.
- **Gowers $k \geq 4$, 2001:** Extended Roth analytic approach by inventing higher-order Fourier analysis and Gowers uniformity norms U^k. Set with no 4-AP must correlate with quadratic phase $e^{2\pi i\left(\alpha n^2+\beta n\right)}$, not just linear. Won Fields Medal.
- **Green-Tao 2004:** Needed Gowers U^3 norm and relative version to handle primes — relative Szemerédi inside pseudorandom majorant.

Why Roth fails for $k \geq 4$:

Count 4-APs:

$$T_4(A) = \sum_{x,d} 1_A(x)1_A(x+d)1_A(x+2d)1_A(x+3d) = N^2 \sum_{r,s} \widehat{1_A}(r)\widehat{1_A}(s)\widehat{1_A}(-2r-s)\widehat{1_A}(r+2s)$$

Four linear phases not enough — quadratic phases $e^{2\pi i \alpha x^2}$ are uniform in U^2 — all linear Fourier coefficients $\left|\hat{f}(r)\right|$ small — but still cause many 4-APs to be missing or overcounted.

Example: $A = \left\{x : \left\|\alpha x^2\right\| < \delta\right\}$ — level set of quadratic phase. Then $\left|\widehat{1_A}(r)\right| = o(N)$ for all $r \neq 0$ — Fourier uniform, looks random to Roth — but $\|1_A\|_{U^3}$ large — contains many 4-APs structure.

So need higher-order uniformity.

Szemerédi 1975:

Proved for all k,

$$\bar{d}(A) > 0 \implies A \text{ contains } k\text{-AP}$$

Equivalently $r_k(N) = o(N)$.

Method: Purely combinatorial, no Fourier. Invented Szemerédi Regularity Lemma — any graph can be partitioned into $O_\epsilon(1)$ pieces where between pieces edges random-like $|e(X,Y) - \delta|X||Y|| < \epsilon|X||Y|$.

Used to build hypergraph regularity for k-AP hypergraph. Proof enormous — $N(\delta, k)$ tower of exponentials of height $O(1/\delta)$ — $\exp^{(O(1/\delta))}(1)$.

Gives no explicit good bound, but qualitative.

Gowers 2001 — Fourier revived:

Invented Gowers uniformity norms:

$$\|f\|_{U^2}^4 = \mathbb{E}_{x,h_1,h_2} f(x)\overline{f(x+h_1)f(x+h_2)}f(x+h_1+h_2)$$

$$\|f\|_{U^3}^8 = \mathbb{E}_{x,h_1,h_2,h_3} \prod_{\omega\in\{0,1\}^3} C^{|\omega|} f(x+\omega\cdot h)$$

In general $\|f\|_{U^k}$ measures correlation with degree $(k-1)$ polynomial phases.

- $\|f\|_{U^2}$ small $\iff$ all $\left|\hat{f}(r)\right|$ small — linear uniform.
- $\|f\|_{U^3}$ small $\iff$ no correlation with quadratic phases $e^{2\pi i(\alpha n^2+\beta n)}$.

Inverse theorem: If $\|f\|_{U^k} \geq \eta$, then f correlates with $(k-1)$-step nilsequence — generalized polynomial phase on nilmanifold.

For $k=3$, nilsequence = linear phase $e^{2\pi irx/N}$ — Roth.

For $k=4$, nilsequence = quadratic phase $e^{2\pi i(\alpha x^2+\beta x)}$ plus bracket polynomials — Gowers.

Proof then: If A no k-AP, balanced f has large U^{k-1} norm → correlates with nilsequence → density increment on Bohr-nil Bohr set → iterate. Gives

$$r_k(N) \ll \frac{N}{(\log\log N)^{c_k}}$$

with explicit $c_k = 2^{-2^{k+9}}$ — first reasonable bound since Szemerédi.

Fields Medal 1998? Actually 1998? No, Gowers Fields 1998 for earlier, but this work core.

Green-Tao 2004 — relative version:

Need Roth/Gowers for primes density 0. Primes $\pi(N) \sim \dfrac{N}{\log N}$ — $\delta(N) = \frac{1}{\log N} \to 0$, so Szemerédi not apply.

Idea: find pseudorandom majorant ν with

1. $0 \leq c \cdot 1_{prime} \leq \nu$,
2. $\mathbb{E}\nu = 1 + o(1)$,
3. $\|\nu - 1\|_{U^{k-1}} = o(1)$ — ν looks like 1 for k-AP counts.

Then prove **relative Szemerédi:** If $0 \leq f \leq \nu$ and $\mathbb{E}f \geq \delta > 0$, then f contains $\geq c(\delta,k)N^2$ k-APs.

Ingredients: generalized von Neumann — if $\|f\|_{U^{k-1}}$ small relative to ν, k-AP count $\approx \delta^k N^2$; dense model theorem — any $0 \leq f \leq \nu$ has model $\tilde{f}$ with $0 \leq \tilde{f} \leq 1$ and $\left\|f - \tilde{f}\right\|_{U^{k-1}} = o(1)$; then apply Gowers inverse to $\tilde{f}$.

For primes, after W-trick $W = \prod_{p\leq w} p$, define

$$\tilde{f}(n) = \frac{\phi(W)}{W} \log N \cdot 1_{prime}(Wn+1)$$

mean ≈ 1, bounded by Selberg sieve ν, so relative Szemerédi gives k-APs in $Wn+1$ primes → k-APs in primes.

Thus Roth seed:

> **If no pattern, then bias, then denser substructure, iterate**

universal template.

In Ramsey terms: $r_3(N)$ inverse of $W(2,3)$ in density form. $W(2,3)=9$ says 2-coloring of $[1,9]$ forces monochrome 3-AP. Roth says even one color class of positive density $\delta>0$ forces it — no need to color rest. Bridge from pigeonhole coloring to analytic density, without it Szemerédi and Green-Tao would not exist.

Behrend's Construction — The Limit of How Far You Can Avoid Order

If Roth's theorem says "dense sets must contain a 3-term AP," Behrend's construction says "you can be *almost* dense and still avoid one." It is the ultimate counterexample — the lower bound that sandwiches $r_3(N)$ and proves Roth's theorem cannot be improved too far.

Let

$$r_3(N)=\max\{|A| : A\subset\{1,\dots,N\},\ A \text{ contains no } a, a+d, a+2d \text{ with } d\neq 0\}.$$

- Roth gives upper bound: $r_3(N)$ cannot be too big, otherwise 3-AP forced.
- Behrend gives lower bound: $r_3(N)$ can be at least this big while still AP-free.

Together:

$$N\cdot\exp(-C\sqrt{\log N})\le r_3(N)\le\frac{N}{(\log N)^{1+c}}$$

For 62 years Behrend's lower bound was unbeaten. It shattered the pre-1946 intuition that AP-free sets must be polynomially small.

The Intuition Before Behrend: Power-Law Conjecture

Erdős and Turán in the 1930s conjectured that any 3-AP-free set must be tiny, like $N^{0.9}$ or $N^{1-\epsilon}$ or even $N/(\log N)^C$ — a power-law saving over N.

Why they thought this: The easy greedy construction — take numbers with no digit 2 in base 3, the Stanley sequence — gives $N^{\log_2 3}\approx N^{0.63}$ and is 3-AP-free. That looks polynomial.

Behrend in 1946 destroyed this. He showed you can keep $N^{1-o(1)}$ numbers — N divided by something growing slower than any N^ϵ — and still avoid 3-APs.

Compare as N grows:

N	POWER LAW GUESS $N^{0.9}$	BEHREND $N\cdot$ EXP($-\sqrt{\text{LOG } N}$)	DENSITY
10^4	3,981	~1,800	18%
10^{10}	**10^9**	**3.1×10^8**	3.1%
10^{20}	**10^{18}**	**6.7×10^{17}**	6.7%
10^{100}	**10^{90}**	**3.8×10^{95}**	**10^{-5}** fraction, but **10^5** times larger than $N^{0.9}$

$\exp(-\sqrt{\log N})$ decays to 0 slower than any $N^{-\epsilon} = \exp(-\epsilon \log N)$, because $\sqrt{\log N} \ll \epsilon \log N$. So Behrend sets are eventually *vastly* larger than any power-law bound. Erdős-Turán power-law conjecture was false.

The Construction: Spheres Have No 3-Term Lines

Behrend's genius was to use geometry.

Key geometric fact: A straight line can intersect a sphere in at most 2 points. If you have three distinct collinear points x, y, z with y midpoint of x and z, they cannot all lie on the same sphere centered at origin, because then $\|x\|^2 = \|y\|^2 = \|z\|^2$ and $\|y\|^2 = \|(x+z)/2\|^2 < (\|x\|^2 + \|z\|^2)/2$ by strict convexity of L^2 norm unless $x = z$.

So a sphere is 3-AP-free.

Step 1: Grid in high dimensions.

Fix d and m. Consider the grid

$$G = \{0, 1, \dots, m-1\}^d \subset \mathbb{Z}^d$$

Size $|G| = m^d$.

For $x \in G$, define $S(x) = \sum_{i=1}^{d} x_i^2 = \|x\|^2$, which ranges from 0 to $d(m-1)^2$.

There are only dm^2 possible values of S, but m^d points. By pigeonhole, some radius R contains many points:

$$\exists R : |\{x \in G : \|x\|^2 = R\}| \geq \frac{m^d}{dm^2} = \frac{m^{d-2}}{d}$$

This set T_R lies on a sphere and thus has no 3-term AP in $\mathbb{Z}^d$ — because if $x + z = 2y$, then x, y, z collinear with y midpoint, impossible on sphere.

Step 2: Map to integers without creating APs.

Encode $x = (x_1, \dots, x_d)$ as a base-$(2m)$ number:

$$\phi(x) = \sum_{i=1}^{d} x_i (2m)^{i-1}$$

Base $2m$ is larger than $2(m-1)$, so addition in $\mathbb{Z}^d$ has no carries when adding two such numbers: $\phi(x) + \phi(z) = 2\phi(y)$ iff $x + z = 2y$ coordinate-wise. Since T_R has no solution to $x + z = 2y$ with $x \neq z$, $\phi(T_R)$ has no 3-term AP in integers.

So $A = \phi(T_R) \subset [0, (2m)^d)$ is AP-free with size $\geq m^{d-2}/d$.

Step 3: Optimize d and m.

We have $N \approx (2m)^d$. So $\log N \approx d \log(2m)$. We have $|A| \geq m^{d-2}/d = N \cdot \dfrac{(2m)^{-2}}{d} \cdot m^d/(2m)^d$... More careful: $m^{d-2}/d = (2m)^d \cdot \dfrac{1}{d(2m)^2 2^d}$? Let's optimize.

Take $m = \exp(\sqrt{\log N})$? Standard optimization: choose $d \approx \sqrt{\log N}$ and $m \approx \exp(\sqrt{\log N})$. Then $(2m)^d = N$ and $m^{d-2}/d = N \cdot \exp(-O(\sqrt{\log N}))$.

Precise bound Behrend proved:

$$r_3(N) \geq N \cdot \exp(-C\sqrt{\log N})$$

for some absolute constant C (original $C = 2\sqrt{2} + o(1)$).

Elkin's Tweak (2008) — The Thick Shell

For 62 years no one beat Behrend. The problem: Behrend uses an *infinitely thin* sphere — only points with *exactly* $\|x\|^2 = R$. Most points in the cube lie near but not exactly on that radius.

Elkin's idea: Use a *thick* spherical shell $R \leq \|x\|^2 \leq R + \delta$ — a doughnut layer. It contains far more points, about δ times more. But now a line *can* intersect a thick shell in 3 points — you lose the sphere property.

Elkin's innovation: Inside the thick shell, most triples that form a 3-AP are still rare. He used probabilistic method and careful counting: pick random subset of shell where you delete one point from each 3-AP inside shell. Since number of 3-APs in shell is much smaller than shell size when δ is small ($\approx \sqrt{d}$), you delete little.

This salvages extra points. Result:

$$r_3(N) \geq C_1 \frac{N \log^{1/4} N}{\exp(2\sqrt{2}\sqrt{\log N})}$$

for $C_1 > 0$. Elkin gained a factor $\approx \sqrt{\log N} = (\log N)^{1/4}$? Actually $\log^{1/4} N$ extra. In his form:

$$r_3(N) \geq \frac{N}{\exp(-C\sqrt{\log N})} \cdot \frac{\sqrt{\log N}}{2^{\cdots}}$$

Green & Wolf (2010) refined to $C = 2\sqrt{2 \log 2}$ improvement.

The improvement is tiny — $\sqrt{\log N}$ vs $\exp(\sqrt{\log N})$ — but conceptually huge: it showed Behrend was not optimal, and thick shells could be made to work.

Why It Defines the Boundary for Roth and Szemerédi

Behrend tells us:

> You cannot prove $r_3(N) \leq N^{0.99}$. You cannot even prove $r_3(N) \leq \frac{N}{(\log N)^{100}}$. Because Behrend sets of size $\frac{N}{\exp(C\sqrt{\log N})}$ are AP-free and larger than $\frac{N}{(\log N)^{100}}$ for large N.

So Roth's theorem $r_3(N) \ll \frac{N}{\log \log N}$ was far from optimal, but Behrend shows you can never push it down to $N^{1-\epsilon}$. The true $r_3(N)$ lives in the strange intermediate regime $N^{1-o(1)}$.

This is why Szemerédi's regularity lemma was necessary. If AP-free sets were small and structured like $N^{0.9}$, you could find APs by simple pigeonhole. Because Behrend showed they can be large, pseudorandom, and sphere-like — looking random but with hidden quadratic structure — Szemerédi needed a decomposition that separates *all* large sets into structured + pseudorandom pieces. The regularity lemma is forced by Behrend-type examples.

In Fourier terms: Behrend sets have *no* large linear Fourier coefficient — they are U^2-uniform — but they *do* have large U^3 quadratic bias (they live on a sphere). This is why Roth's linear Fourier analysis cannot prove $r_3(N) \ll \frac{N}{\exp(\sqrt{\log N})}$; you need quadratic Fourier analysis, which is exactly what Bloom-Sisask and higher-order methods do.

The Behrend-Roth Gap and Bloom-Sisask Closure

For 70 years:

- Lower: $N\exp(-C\sqrt{\log N})$
- Upper: $\frac{N}{\log\log N}$, then $\frac{N\sqrt{\log\log N}}{\sqrt{\log N}}$, then $\frac{N(\log\log N)^5}{\log N}$

Massive gap: $\exp(\sqrt{\log N})$ vs $\log N$.

Behrend says upper bound can never go below $N\exp(-C\sqrt{\log N})$. Roth says it must go to 0.

Bloom & Sisask (2020) closed half the gap: $r_3(N) \ll \frac{N}{(\log N)^{1+c}}$ with $c > 0$. This is first time upper bound is $\frac{N}{(\log N)^{1+c}}$, i.e., smaller than $\frac{N}{\log N}$.

Why $\frac{N}{\log N}$ is threshold? Because $\sum_{n\in A} 1/n$ diverges if $|A \cap [1, N]| \approx \frac{N}{\log N}$ (like primes). Bloom-Sisask implies any 3-AP-free set has convergent reciprocal sum — proving $k = 3$ case of Erdős's conjecture.

Behrend's construction is thus the compass: it tells us how far we *can* hope to push. Until we match $N\exp(-C\sqrt{\log N})$, we haven't finished. The true $r_3(N)$ is conjectured to be closer to Behrend than to Roth — most experts believe $r_3(N) = N\exp(-\Theta(\sqrt{\log N}))$.

In Ramsey language: Behrend is the construction for R_3, like the 5-cycle is for $R(3,3)$. It is the maximal disorder that avoids order, and its size dictates how sophisticated your order-finding tool must be.